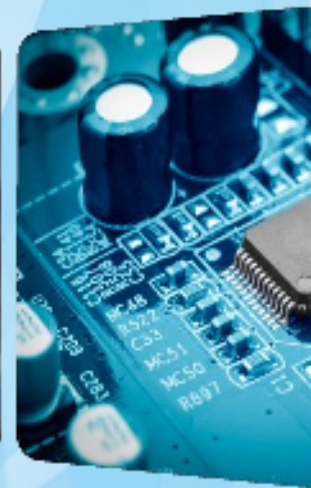




# Shoreline Prediction

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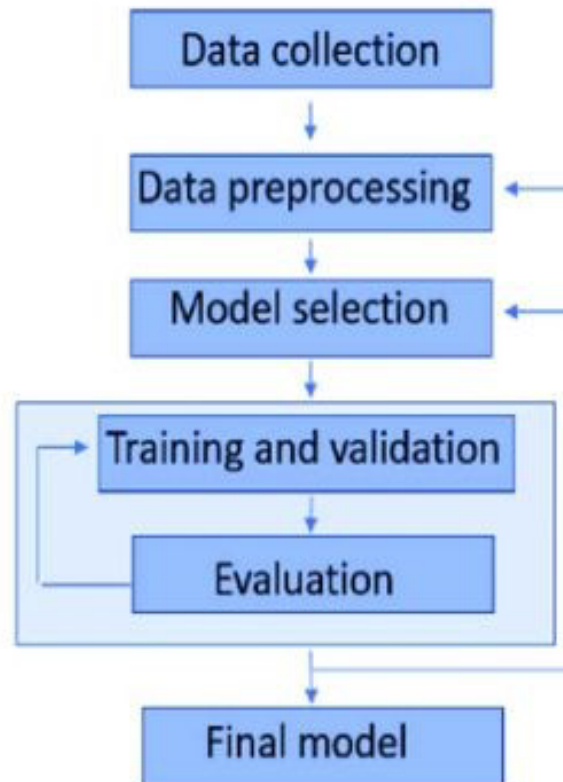
## Problem Statement:

- To analyze and predict the erosion and accretion of soil/sand at the shoreline.
- Predicting the changes that may occur in the specified shoreline across time.

# The Proposed Solution

- Firstly we need to collect the relevant and high resolution images.
- Then we use Qgis to analyze the data set, here we measure the changes that have happened across year over that region.
- Representation and visualization.
- Predictive model using Machine learning.

# The Proposed Solution

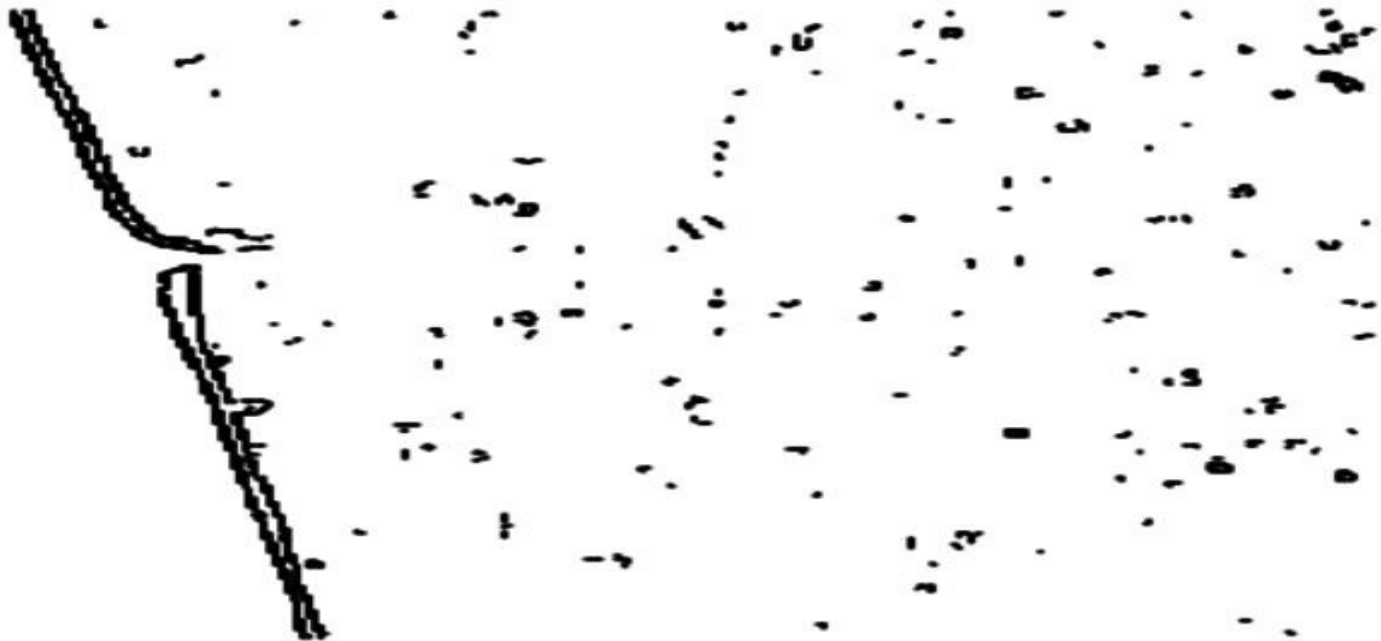


## Raw Dataset (png images):



# Data Preprocessing:

- ▶ We find the edge points, that is the shoreline pixels from a particular dataset.
- ▶ For this purpose we can apply Canny's edge detection algorithm

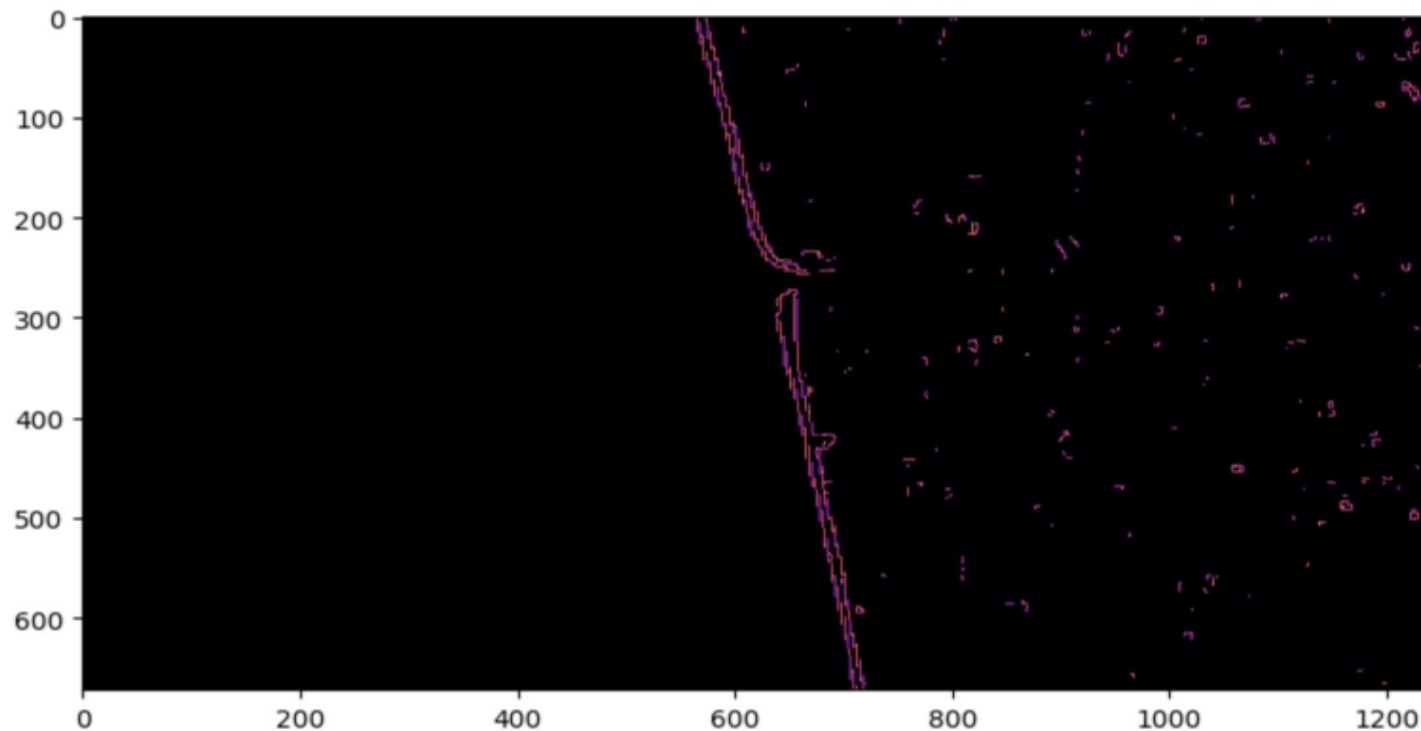


# Canny's Edge Detection

1) In this process we have determined the points where color of ground is changing by this way we have found shoreline

2) There are many unwanted points in this image we have to remove all duplicate points

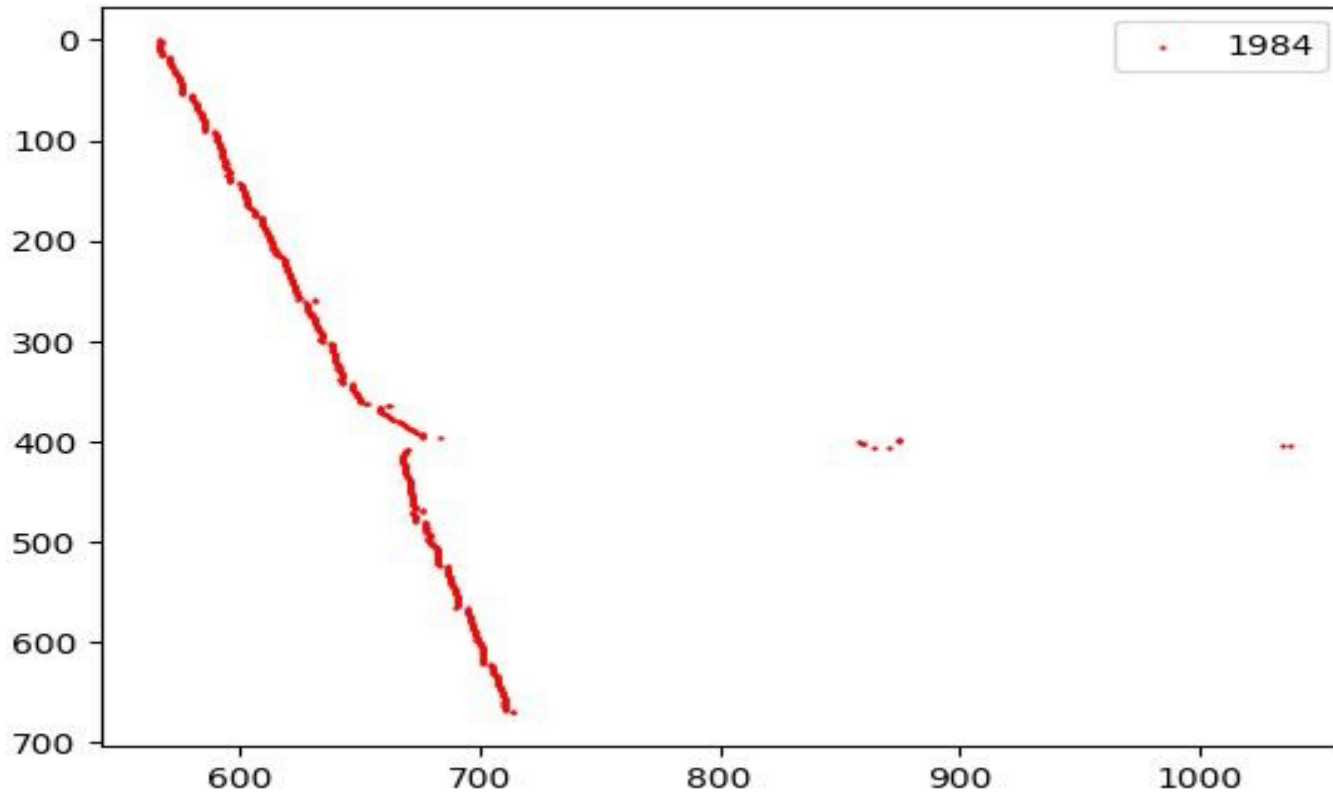
..



## 2. Duplication removal across the horizontal segment

Duplicate points are removed using following piece of code

```
[9]: year=1984
df=pd.DataFrame(DataP)
df.columns=["Axis",year]
df= df.drop_duplicates(subset=["Axis"]).reset_index(drop=True)
```

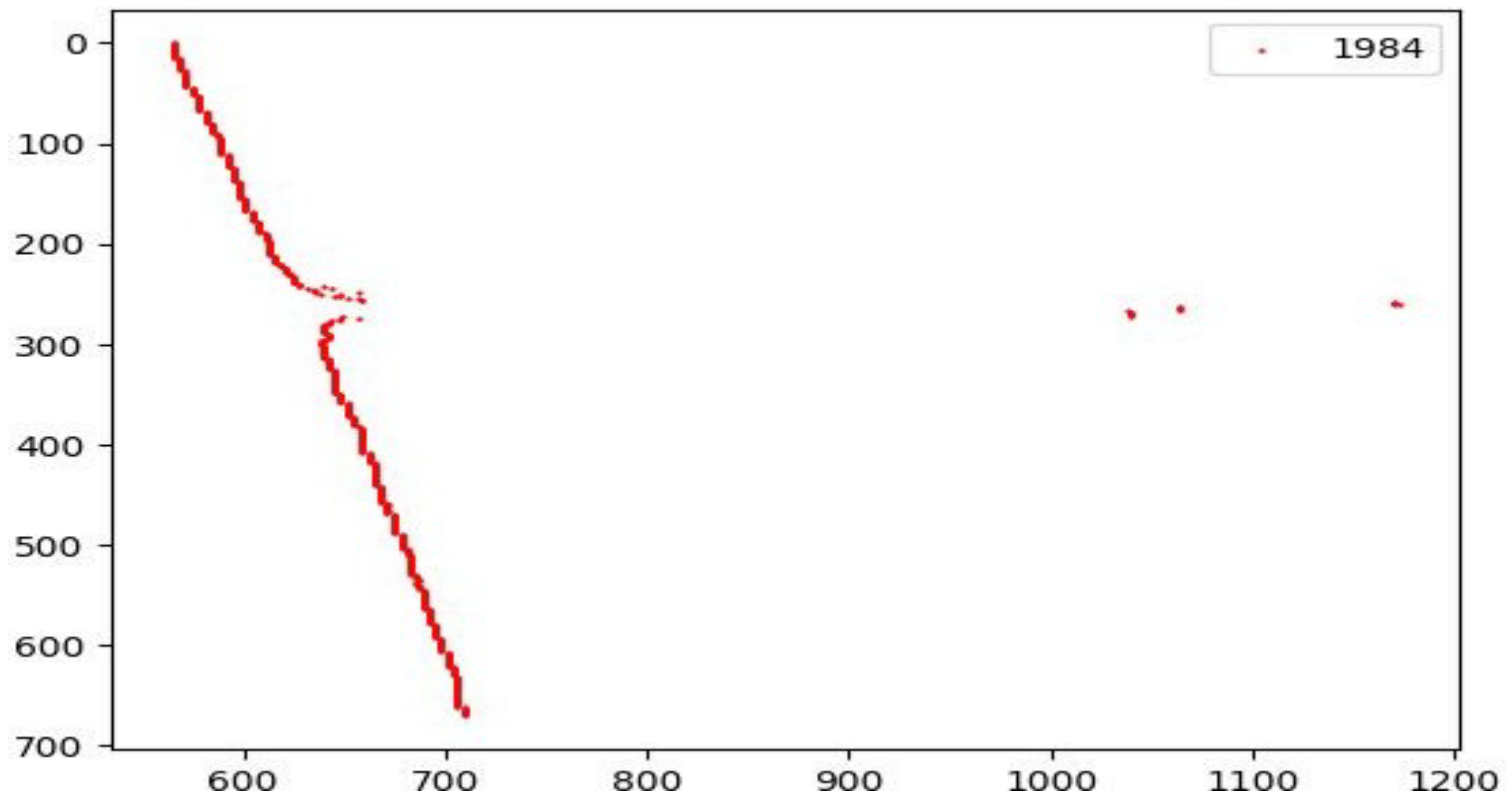


(a) Data points of Shoreline after removing duplicate Points

# Null cell handling

This piece of code removes the null cell cases by replacing the null value with the previous value.

```
#To handle the Null cell cases  
for i in range(1984,2023):  
    c=Points[str(i)]  
    c.fillna(method="ffill",inplace=True)  
    Points[str(i)]=c
```



(b) plot of data points after null cell handling

# Model Building:

This stage of the project involves training various ML models on the pre-processed data and compare their accuracies.

Currently we have built 4 models, which are:

- 1) Linear Regression
- 2) Support Vector Regression (SVR)
- 3) Neural Network
- 4) Decision Tree

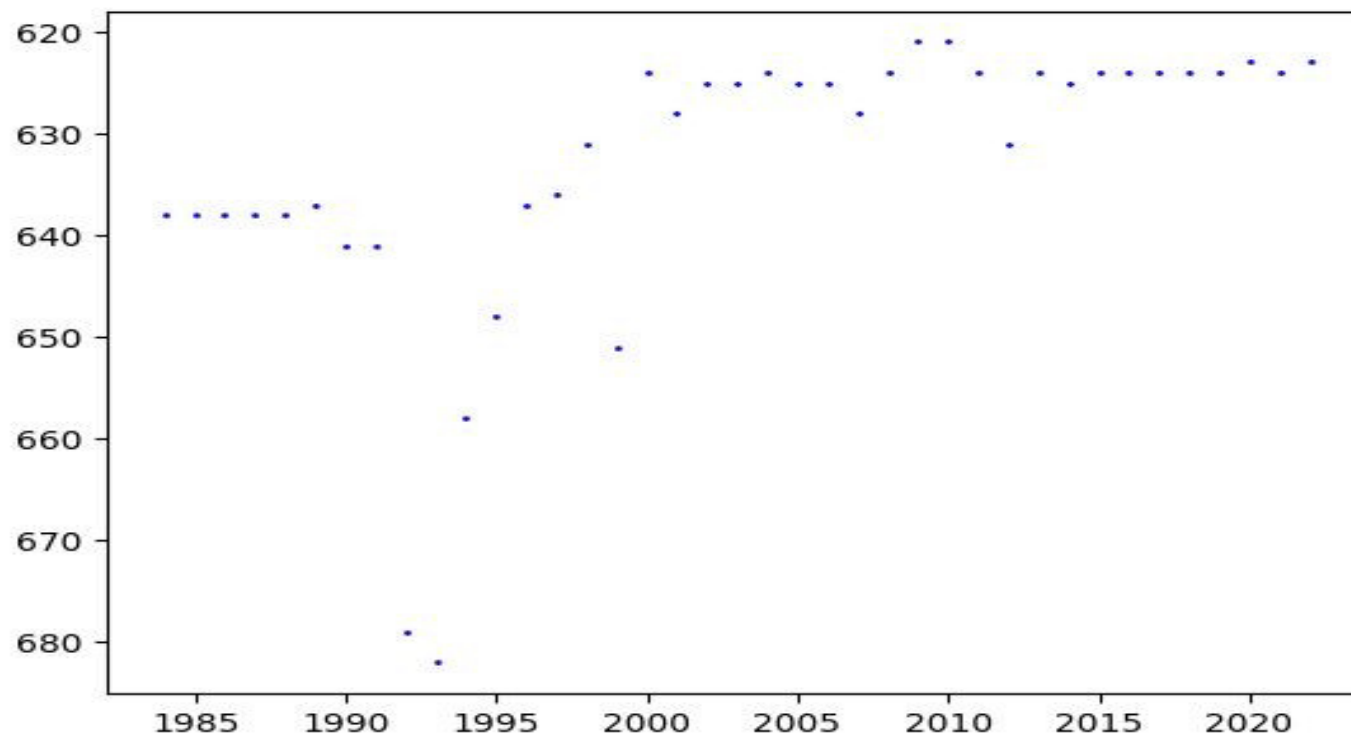
# Model Building

Creation of dataset for training the models is done by implementing the code given below

```
#for one row, a plot on how does the pixel loc change with year  
y = np.array(Points.iloc[250][1:]).reshape(-1, 1)  
years = np.array([i for i in range(1984, 2023)]).reshape(-1, 1)  
  
plt.figure()  
plt.scatter(years,y,s=1,color="blue" )  
plt.gca().invert_yaxis()  
plt.show()
```

# Model Building

The plot of the points of one such dataset.

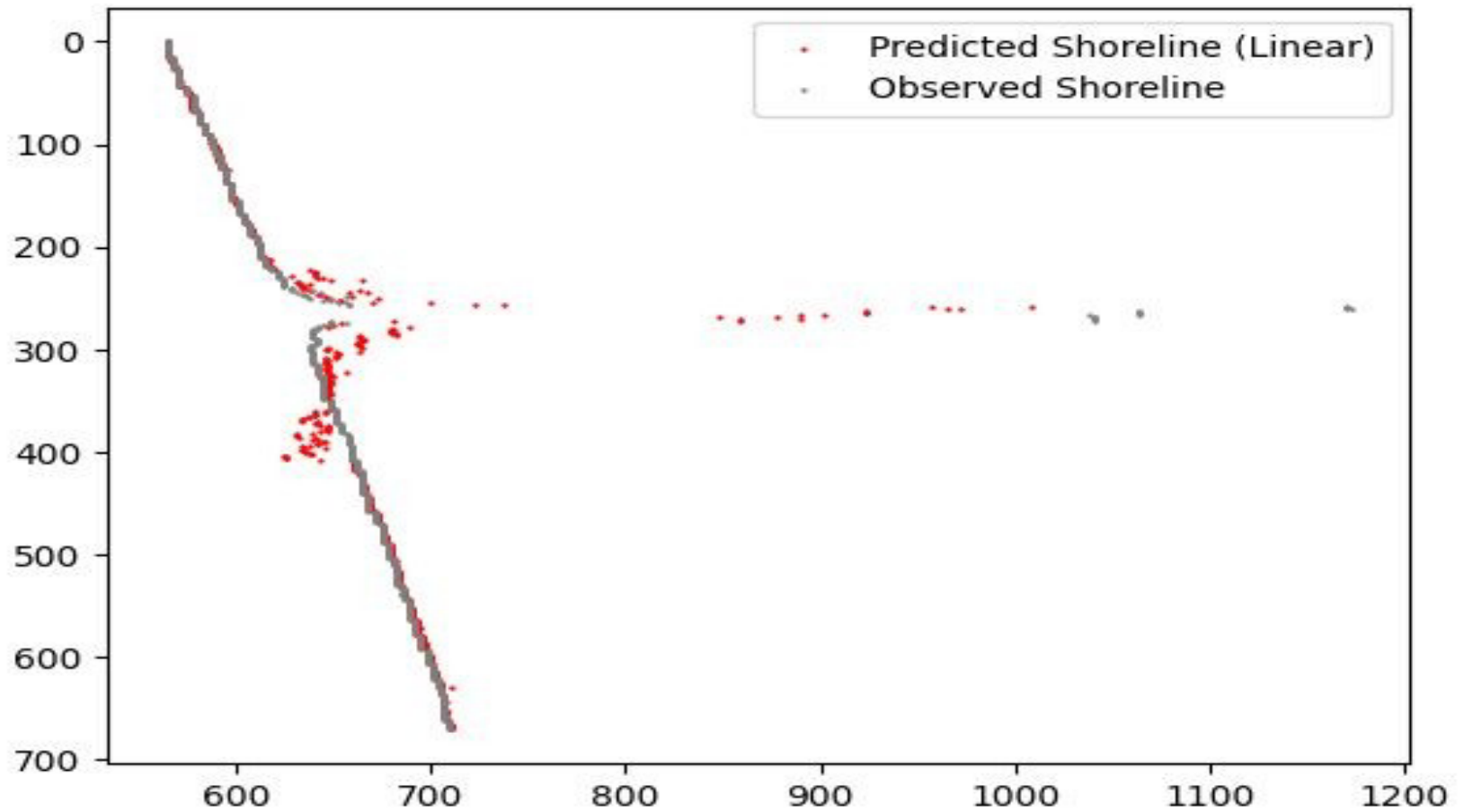


# 1. Linear Regression

Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables.

It's essentially fitting a straight line through a set of data points to capture the overall trend. This involves using a statistical technique to find the line that best fits your data. The most common technique is called least squares, which minimizes the sum of the squared differences between the predicted values and the actual values.

# Linear Regression

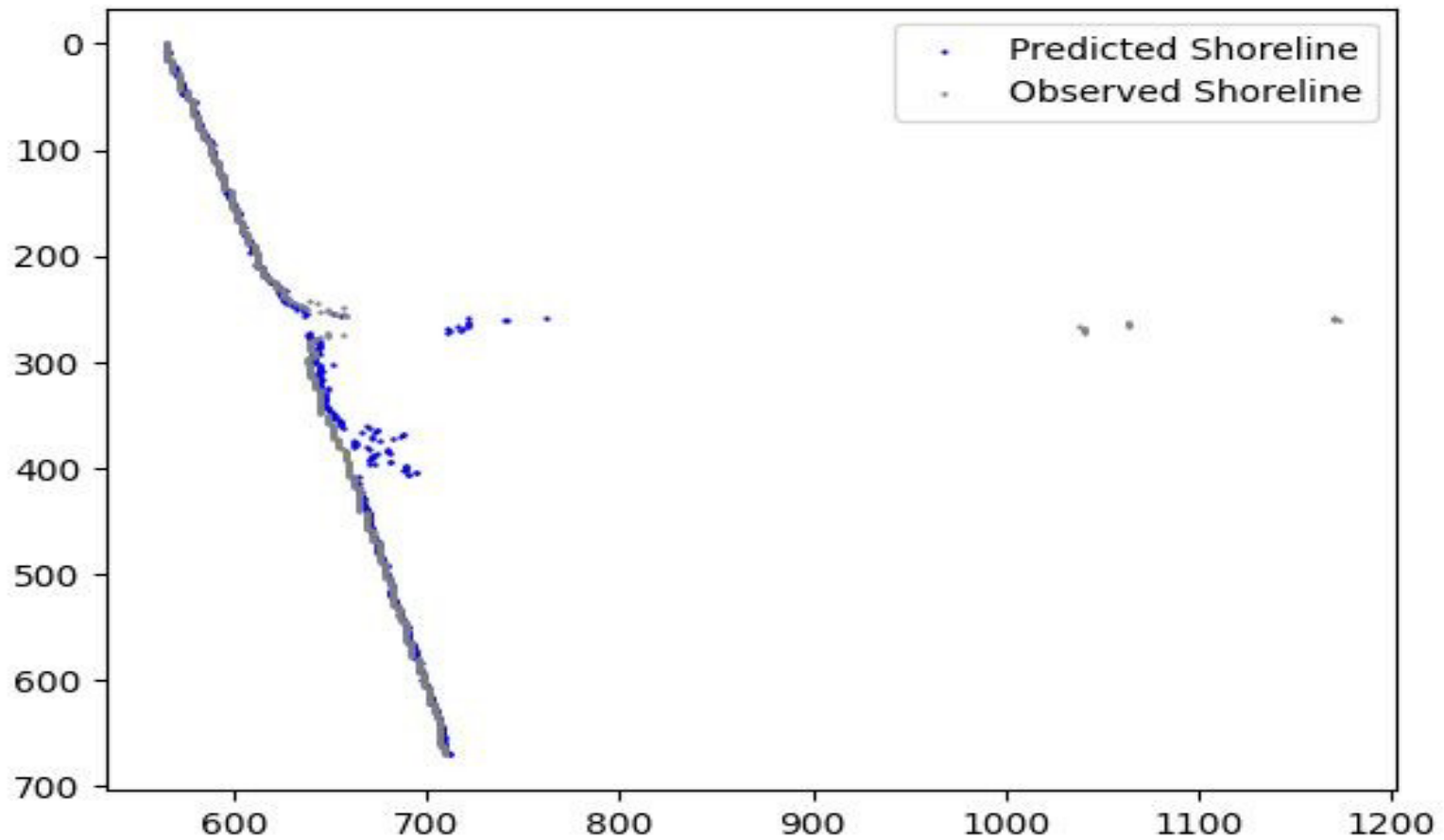


## 2. Support Vector Regression

Support Vector Regression (SVR) is a type of machine learning algorithm used for regression analysis, similar to linear regression.

However, unlike linear regression, which fits a straight line to the data, SVR finds a hyperplane (a high-dimensional plane) that best fits the data while maximizing the margin between the hyperplane and the data points.

# SVR



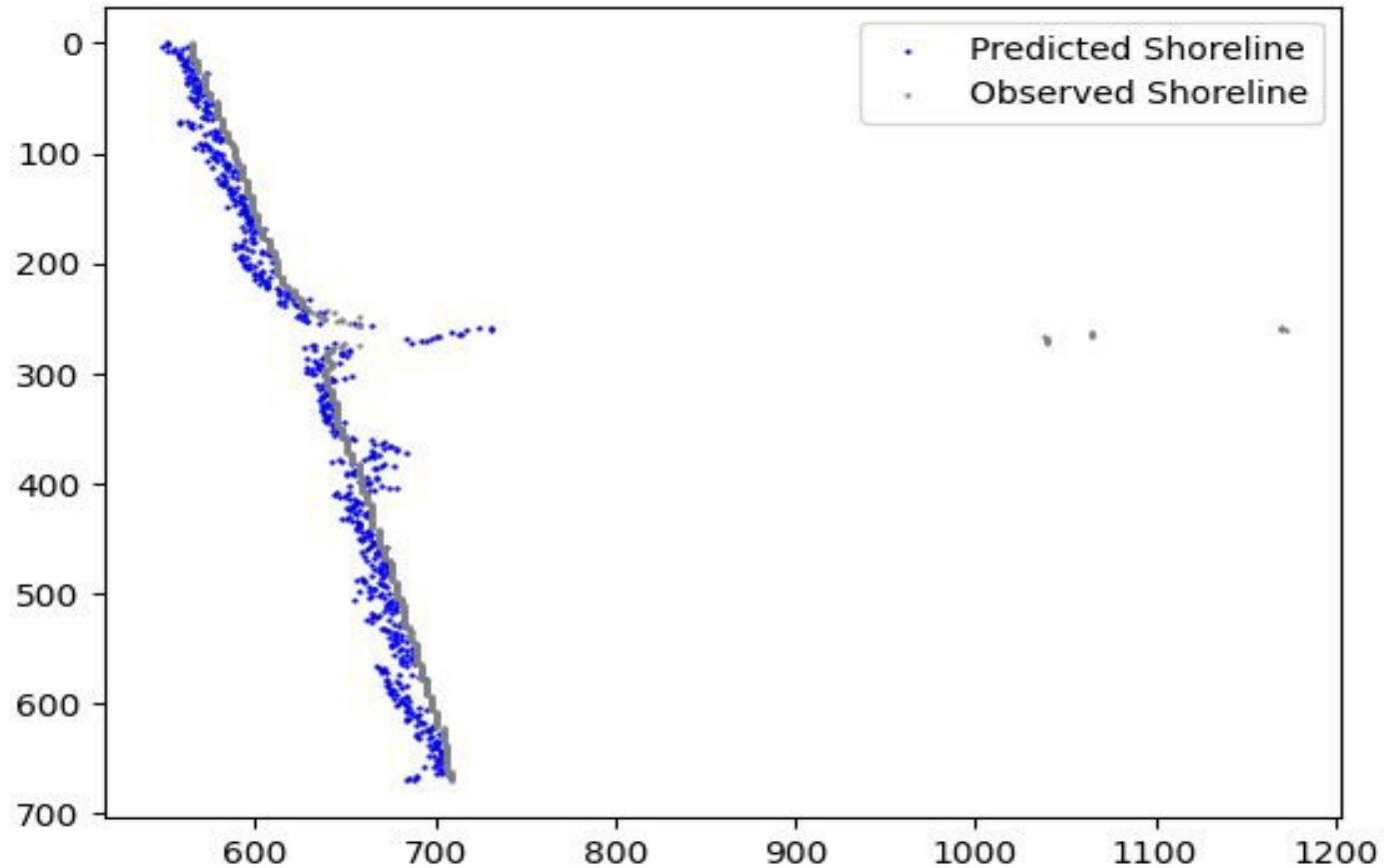
### 3. Neural Network

A neural network, also known as an Artificial Neural Network (ANN), is a complex system inspired by the human brain. It aims to process information and learn from data in a way similar to how our neurons and synapses function.

Training data is fed to the network, consisting of examples with known inputs and desired outputs.

Backpropagation: An algorithm that adjusts the weights based on the difference between the network's actual outputs and the desired outputs. The network "learns" by iteratively optimizing these weights through the training data.

# Neural Network



## 4. Decision Tree

It's a flowchart-like structure where each branch represents a question or test about a specific feature, and each leaf node holds a prediction.

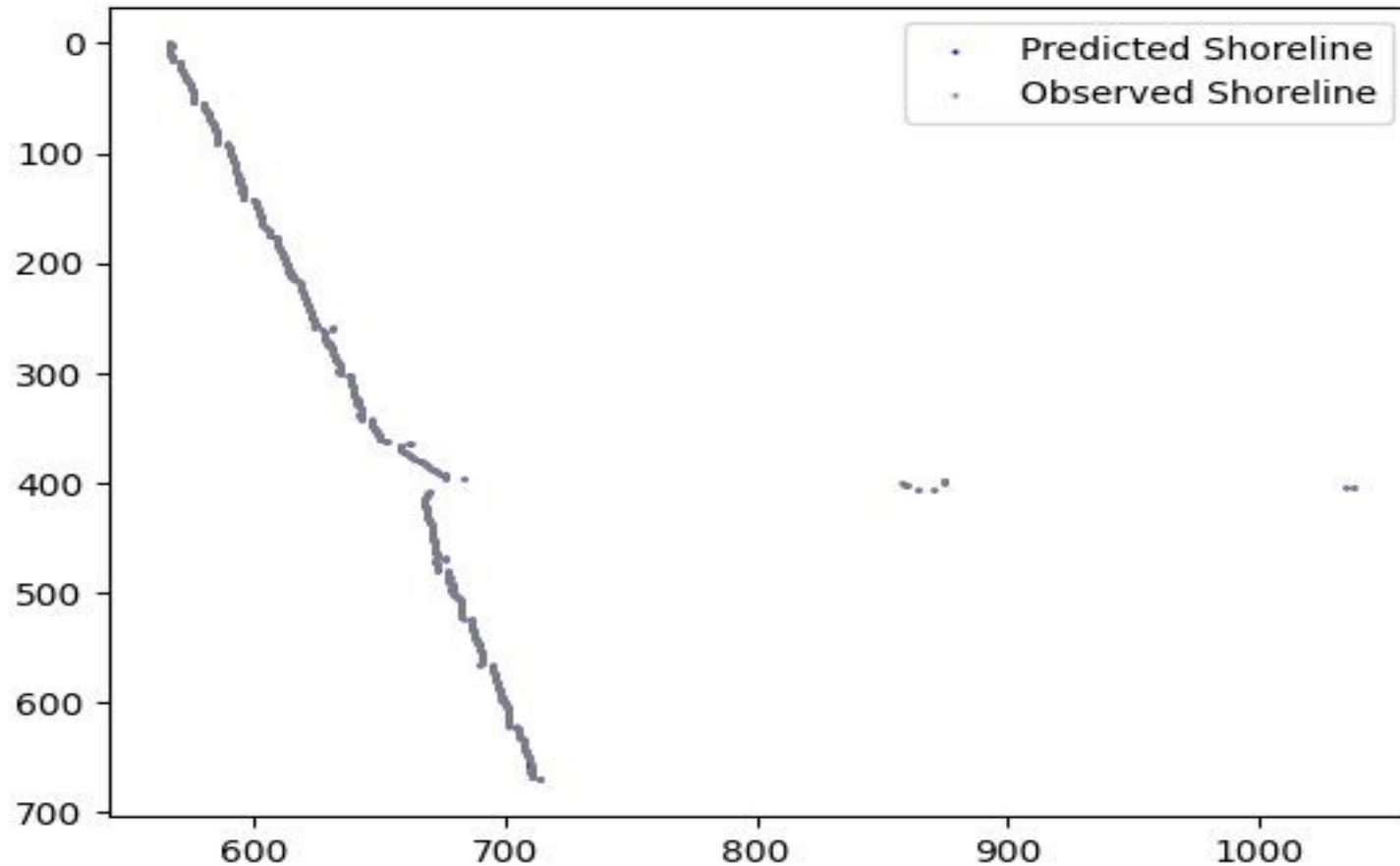
Root node: The starting point, usually representing the main decision or problem.

Branches: Stemming from the nodes, each representing a possible answer or outcome of the previous question.

Internal nodes: Decision points where further questions or tests are made based on specific features.

Leaf nodes: The end points, containing the final prediction or classification for a particular path through the tree.

# Decision Tree



# Thank you!!